Analysis-0 2026 Spring

Homework 5

程昊一

*All of my homework is available at `https://github.com/lacampanella233/Homework.git`.

Problem 1 (3-16). Fix a positive constant α, choose $x_1 > \sqrt{\alpha}$, and define

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right),$$

which determines $x_2, x_3, x_4, \ldots$.

(a) Prove that $\{x_n\}$ is monotonically decreasing and

$$\lim_{n\to\infty} x_n = \sqrt{\alpha}.$$

(b) Let $\varepsilon_n = x_n - \sqrt{\alpha}$. Prove that

$$\varepsilon_{n+1} = \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}.$$

Hence, setting $\beta = 2\sqrt{\alpha}$, obtain

$$\varepsilon_{n+1} < \beta\left(\frac{\varepsilon_1}{\beta}\right)^{2^n}, \qquad n = 1, 2, 3, \ldots.$$

(c) This is a good method for computing square roots, because the recursion is simple and converges very fast. For example, if $\alpha = 3$ and $x_1 = 2$, prove that

$$\frac{\varepsilon_1}{\beta} < \frac{1}{10},$$

and therefore

$$\varepsilon_5 < 4 \cdot 10^{-16}, \qquad \varepsilon_6 < 4 \cdot 10^{-32}.$$

Solution. (a) First, we will show (by induction) that $\sqrt{\alpha} < x_{n+1} < x_n$ for any positive integer n. It suffices to show that if $x_n > \alpha$, then $\alpha < x_{n+1} < x_n$. This is because

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{\alpha}{x_n}\right) > \sqrt{x_n \cdot \frac{\alpha}{x_n}} = \sqrt{\alpha} \text{ (Notice that } x_n \neq \sqrt{\alpha}\text{)},$$

$$x_n - x_{n+1} = \frac{1}{2}\left(x_n - \frac{\alpha}{x_n}\right) = \frac{x_n^2 - \alpha}{2x_n} > 0.$$

Next, we will show that $\lim_{n\to\infty} x_n = \sqrt{\alpha}$. Since $\{x_n\}$ is bounded and monotone, it converges. Let $x = \lim_{n\to\infty} x_n$. Then

$$x = \frac{1}{2}\left(x + \frac{\alpha}{x}\right) \implies x^2 = \alpha \implies x = \sqrt{\alpha}.$$

(b) We have

$$\varepsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2}\left(x_n + \frac{\alpha}{x_n} - 2\sqrt{\alpha}\right) = \frac{1}{2}\left(\sqrt{x_n} - \sqrt{\frac{\alpha}{x_n}}\right)^2 = \frac{\varepsilon_n^2}{2x_n}.$$

Since $x_n > \sqrt{\alpha}$, we have $\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}$. Let $\beta = 2\sqrt{\alpha}$. By induction, we can show that $\varepsilon_{n+1} < \beta\left(\varepsilon_1/\beta\right)^{2^n}$ for any positive integer n. It suffices to show that if $\varepsilon_n < \beta\left(\varepsilon_1/\beta\right)^{2^{n-1}}$, then $\varepsilon_{n+1} < \beta\left(\varepsilon_1/\beta\right)^{2^n}$. This is because

$$\varepsilon_{n+1} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}} < \frac{\beta^2}{2\sqrt{\alpha}}\left(\frac{\varepsilon_1}{\beta}\right)^{2^n} = \beta\left(\frac{\varepsilon_1}{\beta}\right)^{2^n}.$$

(c) Take $\alpha = 3$ and $x_1 = 2$, we can calculate that

$$\varepsilon_1 = x_1 - \sqrt{\alpha} = 2 - \sqrt{3} \approx 0.268,\ \beta = 2\sqrt{3} \approx 3.464 \implies \frac{\varepsilon_1}{\beta} \approx 0.077 < \frac{1}{10}.$$

$\boxed{\textbf{Problem 2}\ \text{(3-17).}}$ Fix $\alpha > 1$. Choose $x_1 > \sqrt{\alpha}$ and define

$$x_{n+1} = \frac{\alpha + x_n}{1 + x_n} = x_n + \frac{\alpha - x_n^2}{1 + x_n}.$$

(a) Prove that

$$x_1 > x_3 > x_5 > \cdots.$$

(b) Prove that

$$x_2 < x_4 < x_6 < \cdots.$$

(c) Prove that

$$\lim_{n\to\infty} x_n = \sqrt{\alpha}.$$

(d) Compare the convergence rate of this method with that of the method in Exercise 16.

Solution. (a) and (b) From the recursion, we have

$$x_{n+2} - x_n = \frac{(1+\alpha)x_n + 2\alpha}{2x_n + (1+\alpha)} - x_n = -\frac{2(x_n^2 - \alpha)}{2x_n + (1+\alpha)} \begin{cases} < 0, & \text{if } x_n > \sqrt{\alpha}, \\ > 0, & \text{if } x_n < \sqrt{\alpha}. \end{cases}$$

$$x_{n+1} - \sqrt{\alpha} = \frac{(\sqrt{\alpha} - 1)(\sqrt{\alpha} - x_n)}{1 + x_n} \begin{cases} < 0, & \text{if } x_n > \sqrt{\alpha}, \\ > 0, & \text{if } x_n < \sqrt{\alpha}. \end{cases}$$

Then the conclusions follow immediately from $x_1 > \sqrt{\alpha}$ and little effort on induction.

(c) Since $\{x_{2n-1}\}$ is bounded and monotone, it converges. Let $x = \lim_{n\to\infty} x_{2n-1}$. Then

$$x = \frac{(1+\alpha)x_n + 2\alpha}{2x_n + (1+\alpha)} \implies x = \pm\sqrt{\alpha}.$$

Since all the terms are positive, we have $x = \sqrt{\alpha}$.

Similarly, we can show that $\lim_{n\to\infty} x_{2n} = \sqrt{\alpha}$. Therefore, $\lim_{n\to\infty} x_n = \sqrt{\alpha}$.

(d) Let $\varepsilon_n = x_n - \sqrt{\alpha}$. We have

$$\varepsilon_{n+1} = \frac{(\sqrt{\alpha}-1)(\sqrt{\alpha}-x_n)}{1+x_n} = \frac{(\sqrt{\alpha}-1)\varepsilon_n}{1+x_n}.$$

Then the convergence rate is linear, i.e. $\varepsilon_n \sim \left(\frac{\alpha-1}{\alpha+1}\right)^n$, which is much slower than the method in Exercise 16.

Problem 3 (3-20). Let $\{p_n\}$ be a Cauchy sequence in a metric space X, and suppose that a subsequence $\{p_{n_i}\}$ converges to a point $p \in X$. Prove that the whole sequence $\{p_n\}$ converges to p.

Proof. For any positive real number ε, there exists integer N_1 and N_2 such that $d(p_{n_i}, p) < \varepsilon/2$ for all $n_i \geq N_1$ and $d(p_m, p_n) < \varepsilon/2$ for all $m, n \geq N_2$.

Take $N = \max\{N_1, N_2\}$. For any $n > N$, select $n_i > n$. Then by the definition of N, we have

$$d(p_n, p) \leq d(p_n, p_{n_i}) + d(p_{n_i}, p) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence, $\{p_n\}$ converges to p. □

Problem 4 (3-21). Prove the following theorem, analogous to Theorem 3.10(b): if, in a complete metric space X, each E_n in the sequence $\{E_n\}$ is a nonempty bounded closed set, $E_n \supset E_{n+1}$, and

$$\lim_{n\to\infty} \operatorname{diam}(E_n) = 0,$$

then

$$\bigcap_{n=1}^{\infty} E_n$$

consists of exactly one point.

Proof. First, we will show that $\bigcap_{n=1}^{\infty} E_n$ is nonempty. For each positive integer n, select $p_n \in E_n$. Since E_1 is bounded and $p_n \in E_n \subset E_1$ for any $n \leq 1$, The sequence $\{p_n\}$ is bounded. Therefore, $\{p_n\}$ has a convergent subsequence, say $\{p_{n_i}\}$. Denote $\lim_{i\to\infty} p_{n_i} = p$.

For any $n \in \mathbb{N}_+$, since $\{p_{n_i} : n_i \geq n\} \subset E_n$, E_n is closed, and $p_{n_i} \to p$, we have $p \in E_n$. Thus, $p \in \bigcap_{n=1}^{\infty} E_n$, and the intersection is nonempty.

Next, we will show that $\bigcap_{n=1}^{\infty} E_n$ consists of exactly one point. Suppose that there are two distinct points $p, q \in \bigcap_{n=1}^{\infty} E_n$. Then for any positive integer n, since $p, q \in E_n$, we have $d(p,q) \leq \operatorname{diam}(E_n)$. Letting $n \to \infty$, we have $d(p,q) = 0$, which is a contradiction. Hence, the intersection consists of exactly one point. □

Problem 5 (3-22). Let X be a complete metric space, and let each G_n in the sequence $\{G_n\}$ be an open dense subset of X. Prove Baire's theorem:

$$\bigcap_{n=1}^{\infty} G_n \neq \varnothing$$

(in fact, this intersection is dense in X).

Proof. We will prove that $G := \bigcap_{n=1}^{\infty} G_n$ is dense in X, i.e. for any $x \in X$ and $r > 0$, $N_r(x) \cap G \neq \varnothing$.

Take $x_1 \in N_r(x)$ (such x_1 exists, since G_1 is dense in X), and $r_1 > 0$ such that $N_{r_1}(x_1) \subset G_1$ (Notice that G_1 is open). Suppose that we have selected $x_1, x_2, \ldots, x_n$ and $r_1, r_2, \ldots, r_n$, such that $N_{r_i}(x_i) \subset G_i$ and $N_{r_i}(x_i) \subset N_{r_{i-1}}(x_{i-1})$. Then we can select $x_{n+1} \in N_{r_n}(x_n) \cap G_{n+1}$ (such x_{n+1} exists, since G_{n+1} is dense in X), and $r_{n+1} > 0$ such that $N_{r_{n+1}}(x_{n+1}) \subset G_{n+1}$ and $N_{r_{n+1}}(x_{n+1}) \subset N_{r_n}(x_n)$.

Then we have a sequence of nested closed balls $\left\{\overline{N}_{r_n}(x_n)\right\}$. Since $r_n \to 0$, we have $\operatorname{diam}(\overline{N}_{r_n}(x_n)) = 2r_n \to 0$. By Problem 21, there exists a unique point $y \in X$ such that $y \in \bigcap_{n=1}^{\infty} \overline{N}_{r_n}(x_n)$. Since $N_{r_n}(x_n) \subset G_n$, we have $y \in G_n$ for any positive integer n. Hence, $y \in G = \bigcap_{n=1}^{\infty} G_n$. Moreover, since $y \in N_r(x)$, we have $N_r(x) \cap G \neq \varnothing$. Therefore, $G = \bigcap_{n=1}^{\infty} G_n$ is dense in X. □

Problem 6 (3-23). Let $\{p_n\}$ and $\{q_n\}$ be two Cauchy sequences in metric space X. Prove that $\{d(p_n, q_n)\}$ converges.

Proof. We'll show that $\{d(p_n, q_n)\}$ is a Cauchy sequence. For any positive real number ε, there exists integer N_1 and N_2 such that $d(p_m, p_n) < \varepsilon/2$ for all $m, n \geq N_1$ and $d(q_m, q_n) < \varepsilon/2$ for all $m, n \geq N_2$.

Take $N = \max\{N_1, N_2\}$. For any $m, n > N$, we have

$$\begin{aligned} d(p_m, q_m) - d(p_n, q_n) &\leq d(p_m, p_n) + d(p_n, q_n) + d(q_n, q_m) - d(p_n, q_n) \\ &= d(p_m, p_n) + d(q_n, q_m) < \varepsilon/2 + \varepsilon/2 = \varepsilon. \\ d(p_m, q_m) - d(p_n, q_n) &\geq -d(p_m, p_n) - d(q_n, q_m) > -\varepsilon/2 - \varepsilon/2 = -\varepsilon. \end{aligned}$$

Thus $|d(p_m, q_m) - d(p_n, q_n)| < \varepsilon$. Hence, $\{d(p_n, q_n)\}$ is a Cauchy sequence, and it converges. □

Problem 7 (3-24). Let X be a metric space.

(a) Two Cauchy sequences $\{p_n\}$ and $\{q_n\}$ in X are called equivalent if

$$\lim_{n \to \infty} d(p_n, q_n) = 0.$$

Prove that this defines an equivalence relation (Definition 2.3).

(b) Let X^* be the set of all equivalence classes obtained in this way. If $P \in X^*$ and $Q \in X^*$, choose $\{p_n\} \in P$ and $\{q_n\} \in Q$, and define

$$\Delta(P, Q) = \lim_{n \to \infty} d(p_n, q_n).$$

By Problem 23, this limit exists. Prove that if $\{p_n\}$ and $\{q_n\}$ are replaced by equivalent sequences, then $\Delta(P, Q)$ is unchanged; hence Δ is a metric on X^*.

(c) Prove that the metric space X^* obtained above is complete.

(d) For each $p \in X$, choose a Cauchy sequence all of whose terms are p; let P_p be the member of X^* containing this sequence. Prove that for all $p, q \in X$,

$$\Delta(P_p, P_q) = d(p, q).$$

In other words, the map $\varphi(p) = P_p$ is an isometry from X into X^*.

(e) Prove that $\varphi(X)$ is dense in X^*, and that $\varphi(X) = X^*$ if and only if X is complete.

Using (d), we may identify X with $\varphi(X)$, and thus regard X as embedded in the complete metric space X^*. The space X^* is called the *completion* of X.

Proof. (a) The reflexivity and symmetry of this relation are obvious. We will show that it is transitive. Suppose that $\{p_n\}$ and $\{q_n\}$ are equivalent, and $\{q_n\}$ and $\{r_n\}$ are equivalent. For any positive real number ε, there exists integer N_1 and N_2 such that $d(p_n, q_n) < \varepsilon/2$ for all $n \geq N_1$ and $d(q_n, r_n) < \varepsilon/2$ for all $n \geq N_2$.

Take $N = \max\{N_1, N_2\}$. For any $n > N$, we have

$$d(p_n, r_n) \leq d(p_n, q_n) + d(q_n, r_n) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

Hence, $\{p_n\}$ and $\{r_n\}$ are equivalent, and the relation is transitive.

(b) Assume that $\{p'_n\}$ and $\{q'_n\}$ are equivalent to $\{p_n\}$ and $\{q_n\}$ respectively. Denote $D := \lim_{n\to\infty} d(p_n, q_n)$.

For any positive number ε, there exist N_1, N_2, N_3 such that $|d(p_n, q_n) - D| < \varepsilon/3$ for any $n > N_3$, and $|d(p_n, p'_n)| < \varepsilon/3$ for any $n > N_1$, and $|d(q_n, q'_n)| < \varepsilon/3$ for any $n > N_2$.

Let $N = \max\{N_1, N_2, N_3\}$. Then, for any $n > N$, we have

$$\begin{aligned} |d(p'_n, q'_n) - D| &= |(d(p'_n, q'_n) - d(p_n, q_n))| + |d(p_n, q_n) - D| \\ &\leq |d(p'_n, q'_n) - d(p_n, q_n)| + \varepsilon/3 \\ &\leq |d(p'_n, q'_n) - d(p_n, q'_n)| + |d(p_n, q'_n) - d(p_n, q_n)| + \varepsilon/3 \\ &\leq d(p_n, p'_n) + d(q_n, q'_n) + \varepsilon/3 < \varepsilon/3 + \varepsilon/3 + \varepsilon/3 = \varepsilon. \end{aligned}$$

Thus $|d(p'_n, q'_n) - D| < \varepsilon$ for any $n > N$. Hence, $\lim_{n\to\infty} d(p'_n, q'_n) = D$, and $\Delta(P, Q)$ is unchanged.

(c) Take a Cauchy sequence $\{P_n\}$ in X^*, where $P_n = \{p_{n,m}\}_{m\geq 1}$. Construct $P = \{p_{k,n_k}\}_{k\geq 1}$ as follow:

- Take n_1 such that for any $i, j \geq n_1$, $d(p_{1,i}, p_{1,j}) < 1$.
- Take $n_2 > n_1$ such that: (a) for any $i, j \geq n_2$, $d(p_{2,i}, p_{2,j}) < 1/2$; (b) for any $i \geq n_2$, $d(p_{1,i}, p_{2,i}) < 2\Delta(P_1, P_2)$.
- ...
- If $n_1, \cdots, n_k$ has been taken, then take n_{k+1} such that: (a) for any $i, j \geq n_{k+1}$, $d(p_{k+1,i}, p_{k+1,j}) < 1/(k+1)$; (b) for any $i \geq n_{k+1}$ and $1 \leq j \leq k$, $d(p_{j,i}, p_{k+1,i}) < 2\Delta(P_j, P_{k+1})$.

We will first show that $P \in X^*$. For any positive number ε, take N_1 such that fora any $i, j > N_1$, $\Delta(P_i, P_j) < \varepsilon/4$. Then, let $N = \max\{\lceil 2/\varepsilon \rceil, N_1\}$. For any $i, j > N$, by definition, we have

$$d(p_{i,n_i}, p_{i,n_j}) < \frac{1}{N} < \frac{\varepsilon}{2};\ d(p_{i,n_j}, p_{j,n_j}) < 2\Delta(P_i, P_j) < \frac{\varepsilon}{2}$$
$$\implies d(p_{i,n_i}, p_{j,n_j}) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Hence, P is a Cauchy sequence, and $P \in X^*$.

Next, we will show that $P_n \to P$ in X^*. For any positive number ε, take N_1 such that for any $i, j > N_1$, $d(p_{i,n_i}, p_{j,n_j}) < \varepsilon/4$. Then, let $N = \max\{\lceil 4/\varepsilon \rceil, N_1\}$. For any $m > N$, we will show that $\Delta(P_n, P) < \varepsilon$. For any $i > n_m$,

$$d(p_{m,i}, p_{m,n_m}) < \frac{1}{m} < \frac{\varepsilon}{4},\ d(p_{m,n_m}, p_{i,n_i}) < \varepsilon/4$$
$$\implies d(p_{m,i}, p_{i,n_i}) < \varepsilon/2 \implies \Delta(P_n, P) \leq \frac{\varepsilon}{2} < \varepsilon.$$

Hence, $\lim_{n\to\infty} \Delta(P_n, P) = 0$, and X^* is complete.

(d) The statement can be easily proved by definition.

(e) We will first show that $\varphi(X)$ is dense in X^*, i.e. for any $P = \{p_n\}_{n\geq 1} \in X^*$ and $\varepsilon > 0$, there exists $p \in X$ such that $\Delta(P, P_p) < \varepsilon$. There exists N such that for any $i, j \geq N$, $d(p_i, p_j) < \varepsilon/2$. Let $p = p_N$. Then, for any $i \geq N$, we have

$$d(p_i, p) < \varepsilon/2 \implies \Delta(P, P_p) \leq \varepsilon/2 < \varepsilon.$$

Hence, $\varphi(X)$ is dense in X^*.

Next, we will show that $\varphi(X) = X^*$ if X is complete. For any $P = \{p_n\}_{n\geq 1} \in X^*$, there exists $p \in X$ such that $p_n \to p$. Then P is equivalent to P_p, and $P = P_p$. Hence, $\varphi(X) = X^*$.

□